SOFTWARE REQUIREMENTS SPECIFICATION

Student Attendance Management System

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Including UML Design Diagrams

Use Case • Activity • Class • Sequence • State • Deployment • Component

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1. Introduction

1.1 Purpose

The purpose of the Student Attendance Management System (SAMS) is to automate and simplify the process of recording, managing, monitoring, calculating, and reporting student attendance. The system provides role-based access for administrators, teachers, and students.

1.2 Scope

The system covers authentication, student and teacher management, subject and class management, attendance recording and modification, attendance percentage calculation, attendance reports, attendance history, low-attendance alerts, and search/filtering. The system is suitable for a web-based academic environment.

1.3 Intended Users

- **Administrator:** Manages users, students, teachers, subjects, classes, and reports.
- **Teacher:** Records and manages attendance and generates attendance reports.
- **Student:** Views personal attendance, history, and attendance percentage.

1.4 Definitions and Abbreviations

Term	Definition
SRS	Software Requirements Specification
SAMS	Student Attendance Management System
Admin	System administrator
Attendance	Record indicating whether a student was present or absent
DB	Database
UI	User Interface

2. Overall Description

2.1 Product Perspective

SAMS is an independent application intended to replace or supplement manual attendance processes. Attendance information is maintained in a centralized database. Authorized users access the system through a secure login interface. **Basic flow:** User → Login → Role-based Dashboard → Functional Module → Database

2.2 Product Functions

- User authentication and role-based authorization.
- Student, teacher, subject, class, and section management.
- Teacher attendance recording and authorized modification.
- Automatic attendance percentage calculation.
- Attendance history and report generation.
- Student access to personal attendance information.
- Low-attendance identification and notification.
- Search and filtering of attendance records.

3. Functional Requirements

FR-01: User Authentication

- The system shall authenticate users using valid credentials.

- The system shall identify the user's role and provide role-specific access.
- Invalid credentials shall generate an appropriate error message.
- Users shall be able to log out securely.

FR-02: Student Management

- Administrators shall be able to add, view, update, deactivate, delete, and search student records.
- Student records may contain student ID, name, date of birth, gender, email, phone, address, course, year, semester, andclass/section.

FR-03: Teacher Management

- Administrators shall be able to add, view, update, deactivate, and manage teacher records.
- Teachers may be assigned to subjects and classes.

FR-04: Subject Management

- Administrators shall be able to create and manage subjects including subject ID, code, name, department, semester, credits, and assigned teacher.

FR-05: Class and Section Management

- Administrators shall be able to create classes and sections, assign students to classes, assign subjects to classes, and assign teachers to classes.

FR-06: Attendance Recording

- Teachers shall select class, subject, and date, view the student list, mark Present or Absent, and submit attendance.
- The system shall store attendance records in the database.

FR-07: Attendance Validation

- The system shall prevent duplicate attendance records for the same student, subject, class, date, and session.

FR-08: Attendance Modification

- Authorized teachers or administrators shall be able to modify attendance.
- The system should record original status, updated status, time of modification, and modifying user.

FR-09: Attendance Calculation

- The system shall calculate attendance using: (Classes Attended / Classes Conducted) × 100.
- Attendance shall be available subject-wise, student-wise, class-wise, and semester-wise.

FR-10: Attendance Reports

- The system shall generate student, class, and teacher attendance reports.
- Reports may be displayed and optionally exported as PDF or CSV.

FR-11: Student Attendance View

- Students shall be able to view their own total classes, attended classes, absent classes, percentage, subject-wiseattendance, and history.

FR-12: Low Attendance Notification

- The system shall identify students below a configurable minimum attendance threshold and display a warning.

FR-13: Search and Filtering

- The system shall support searching by student ID, student name, class, section, subject, teacher, date, and attendancestatus.

FR-14: Dashboard

- The system shall provide role-specific dashboards for administrators, teachers, and students.

4. Non-Functional Requirements

NFR-01: Performance

The system should respond to normal requests within a reasonable time and generate reports efficiently.

NFR-02: Security

The system shall use authentication, role-based access control, secure password hashing, and protection against unauthorized access or modification.

NFR-03: Reliability

Attendance data shall be stored reliably. Regular database backups should be maintained.

NFR-04: Usability

The interface should be simple, consistent, readable, and easy to navigate.

NFR-05: Maintainability

The application should use modular components so that individual features can be updated independently.

NFR-06: Scalability

The system should support growth in users, classes, subjects, and attendance records.

NFR-07: Availability

The system should be available during institutional working hours and minimize downtime.

NFR-08: Data Integrity

Unique identifiers, validation, duplicate prevention, and accurate percentage calculations shall be maintained.

5. System Requirements

5.1 Hardware Requirements

- **Server:** Intel Core i3 or higher, minimum 4 GB RAM, minimum 20 GB storage, and network connectivity.
- **Client:** Dual-core processor or higher, minimum 2 GB RAM, minimum 5 GB available storage, keyboard/mouse, and network connection.

5.2 Software Requirements

- **Frontend:** HTML, CSS, JavaScript, Bootstrap or equivalent.
- **Backend:** Java, Python, PHP, Node.js, C#, or another suitable server-side technology.
- **Database:** MySQL, PostgreSQL, Oracle Database, SQL Server, or equivalent.
- **Operating System:** Windows, Linux, or macOS.

6. External Interface Requirements

6.1 User Interface

- Login Page
- Administrator Dashboard
- Teacher Dashboard
- Student Dashboard

- Student Management
- Teacher Management
- Subject Management
- Class Management
- Attendance Entry
- Attendance History
- Attendance Reports
- Profile Page

6.2 Database Interface

The application shall store and retrieve user, student, teacher, subject, class, and attendance information from the database.

6.3 Communication Interface

For a web-based implementation, client and server shall communicate through HTTP/HTTPS. HTTPS should be used to protect authentication and sensitive data.

7. Database Requirements

Users

Field	Description
user_id	Unique user identifier
username	Login username
password_hash	Hashed password
role	Admin/Teacher/Student
status	Active/Inactive

Students

Field	Description
student_id	Unique student ID
name	Student name
email	Student email
phone	Phone number
course	Course
year	Academic year
semester	Semester
class_id	Class reference

Teachers

Field	Description
teacher_id	Unique teacher ID
Field	**Description**
name	Teacher name
email	Teacher email
phone	Phone number
department	Department

Subjects

Field	Description

subject_id	Unique subject ID
subject_code	Subject code
subject_name	Subject name
credits	Subject credits
semester	Semester

Classes

Field	Description
class_id	Unique class ID
class_name	Class name
section	Section
academic_year	Academic year

Attendance

Field	Description
attendance_id	Unique attendance ID
student_id	Student reference
subject_id	Subject reference
teacher_id	Teacher reference
class_id	Class reference
date	Attendance date
status	Present/Absent
marked_at	Date/time recorded

8. Use Case Requirements

8.1 Actors

- Administrator
- Teacher
- Student

8.2 Main Use Cases

- **Administrator:** Login, manage students, teachers, subjects, classes, users, and reports.
- **Teacher:** Login, view assigned classes, mark attendance, edit authorized attendance, view attendance, and generate reports.
- **Student:** Login, view profile, view attendance, view percentage, and view notifications.

8.3 Use Case: Mark Attendance

Actor: Teacher

Precondition: Teacher is logged in and assigned to the selected class and subject.

- Open attendance module
- Select class and subject
- Select attendance date
- View student list
- Mark Present or Absent
- Submit attendance
- System validates data

- System stores attendance
- System displays confirmation

Alternative flow: If attendance already exists for the selected date and subject, the system displays a warning and prevents duplicate entry.

Postcondition: Attendance is stored successfully.

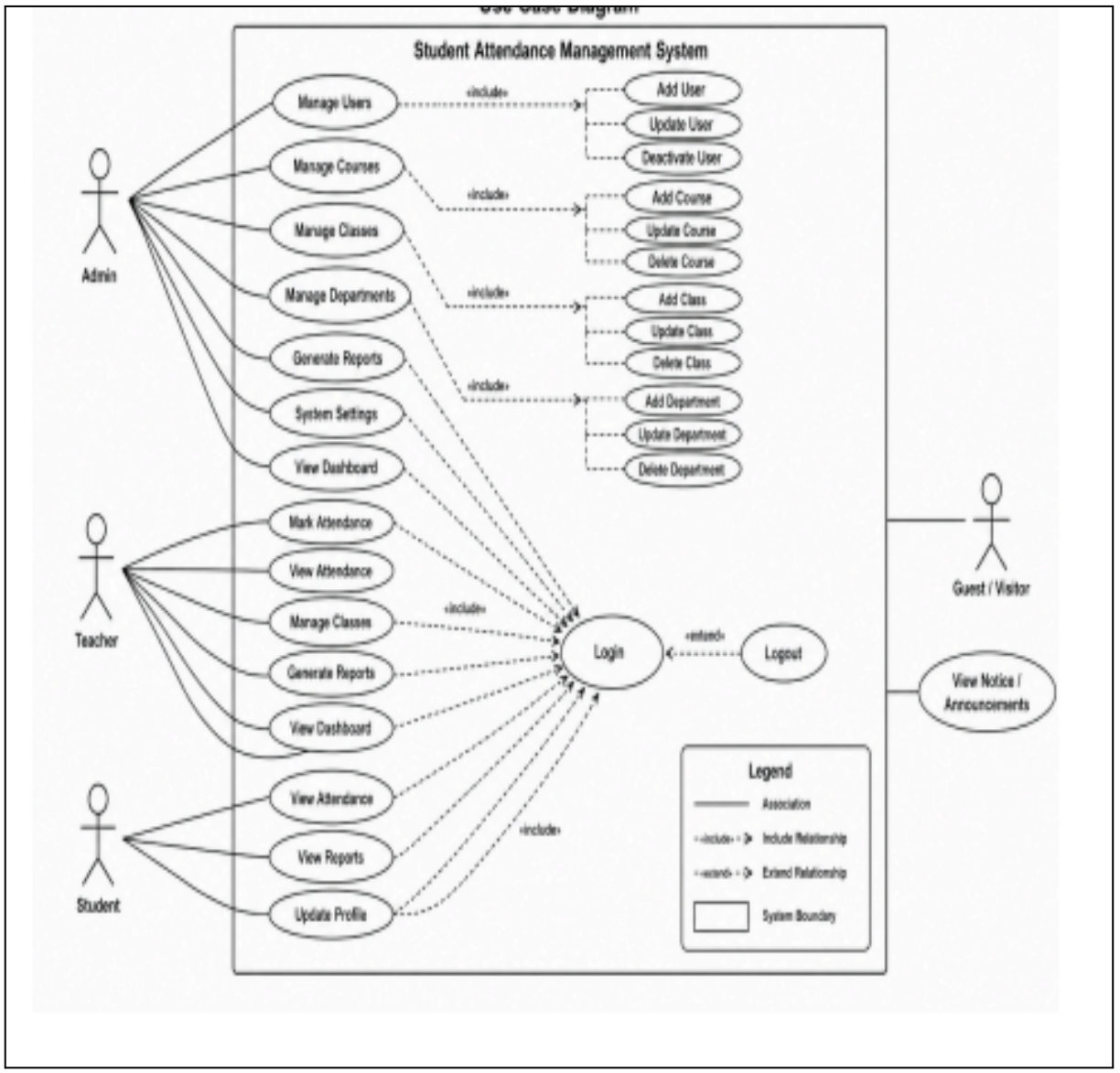
9. UML Design and Diagrams

The following UML diagrams describe the structure, behavior, interactions, lifecycle, physical deployment, and software components of the Student Attendance Management System.

9.1 Use Case Diagram

Shows the interactions between Student, Teacher, Administrator, and the major system functions.

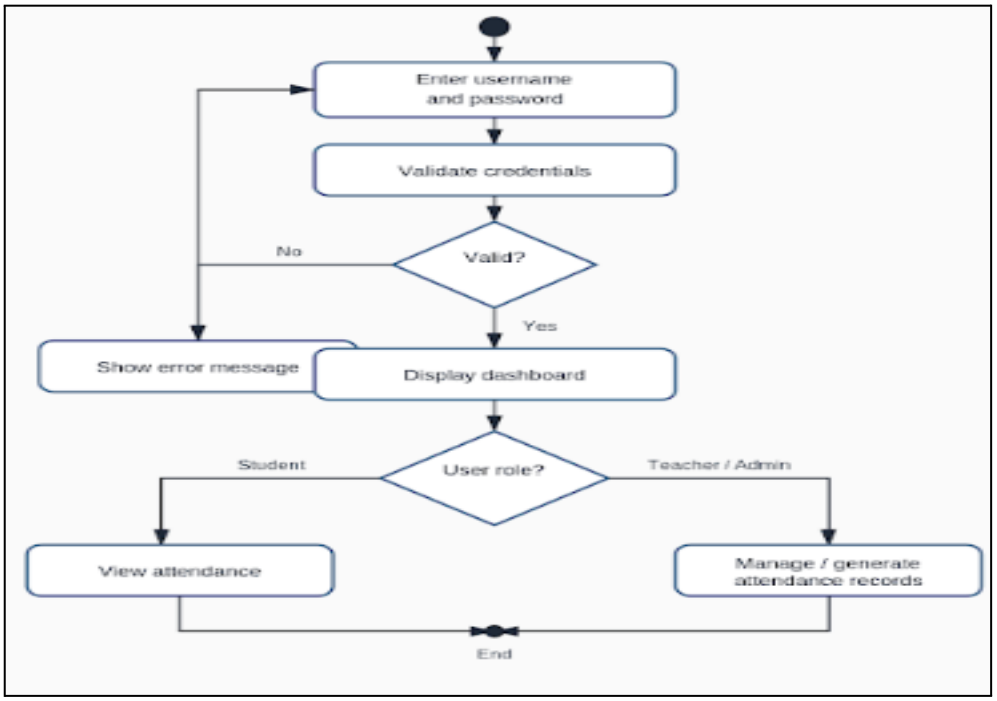
Use Case Diagram



9.2 Activity Diagram

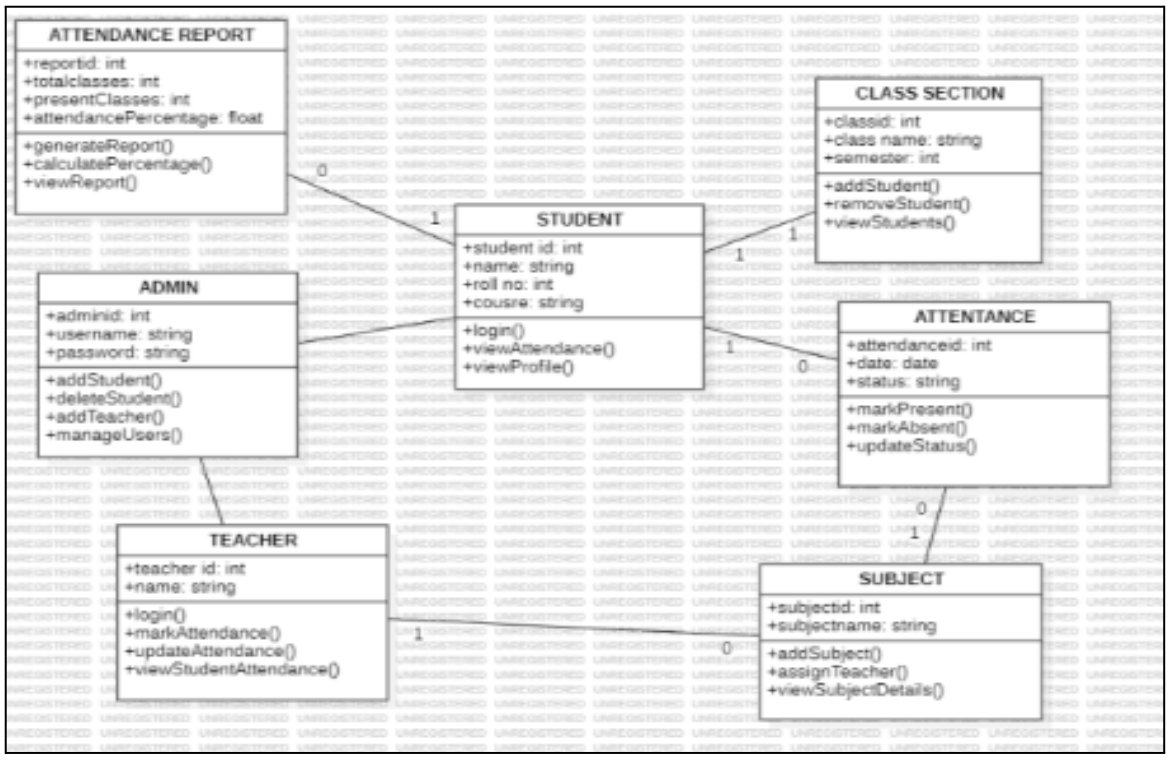
Shows the workflow for marking attendance from login through validation and storage.

Activity Diagram — Mark Attendance



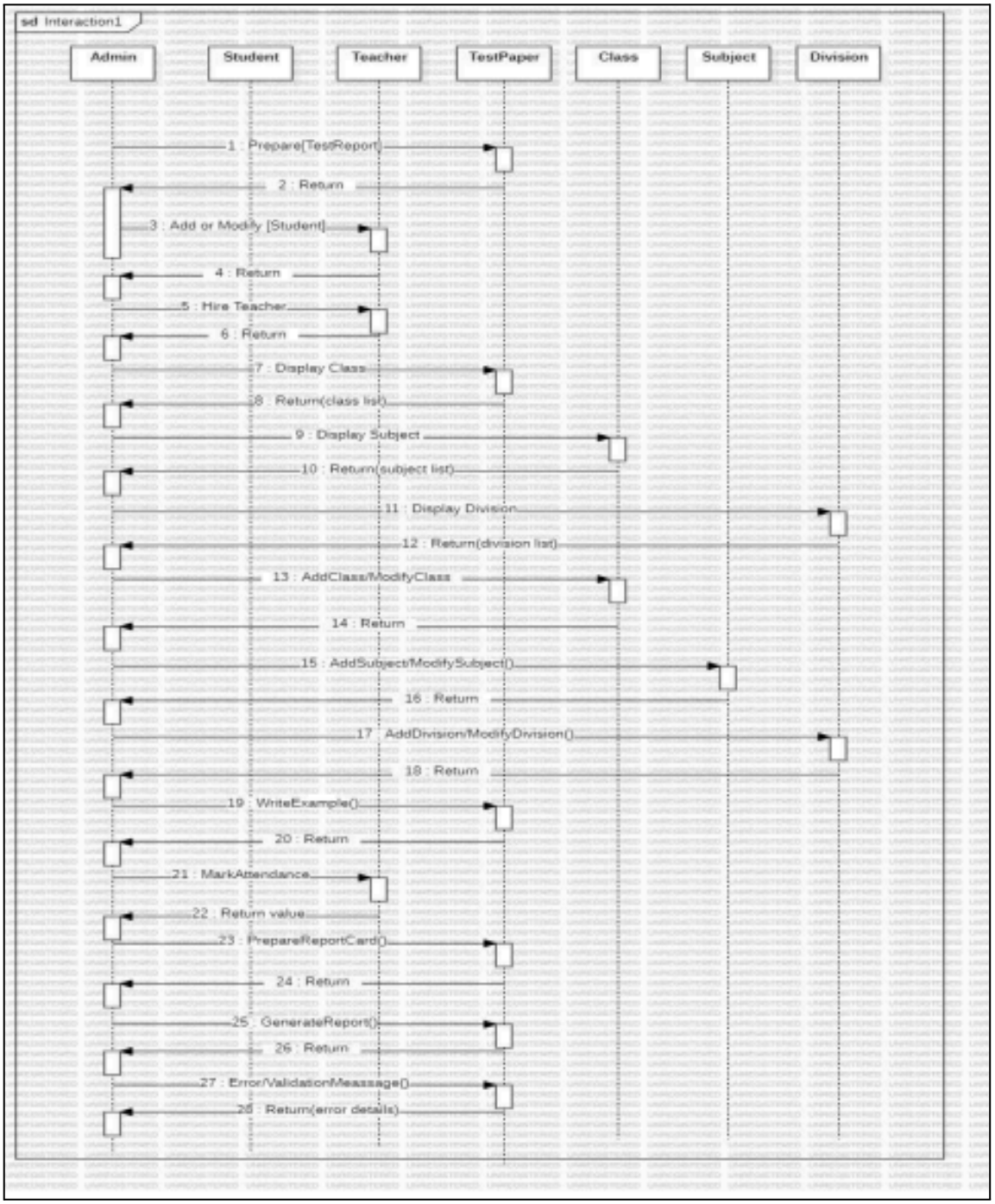
9.3 Class Diagram

Shows major classes, attributes, operations, and relationships in the system.



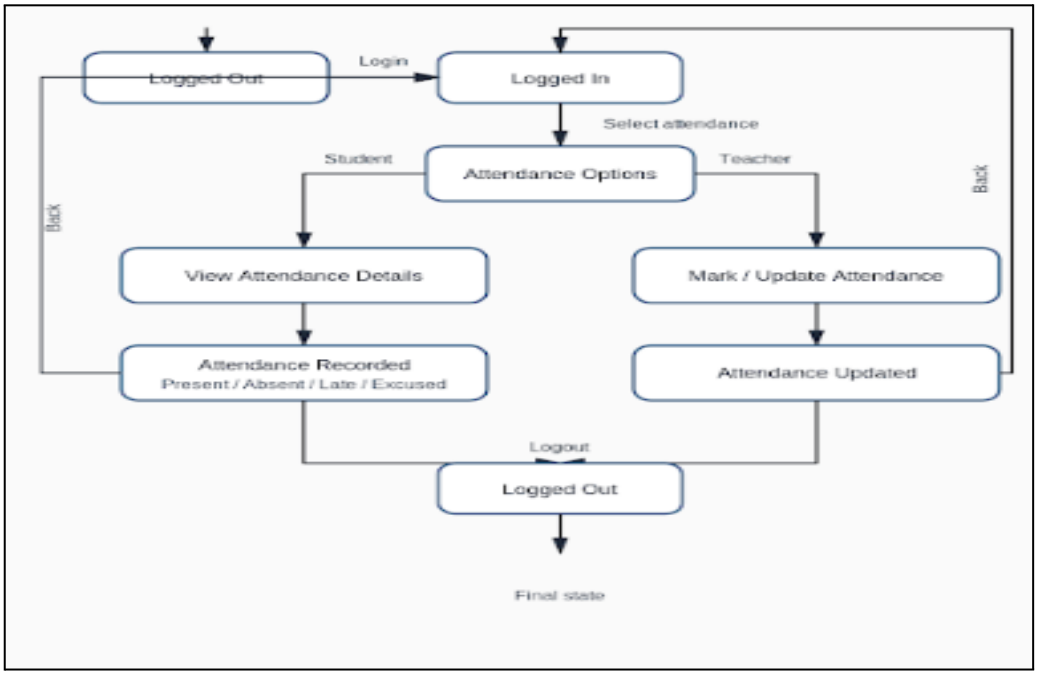
9.4 Sequence Diagram

Shows the time-ordered interaction among Teacher, UI, Attendance Service, and Database while marking attendance.



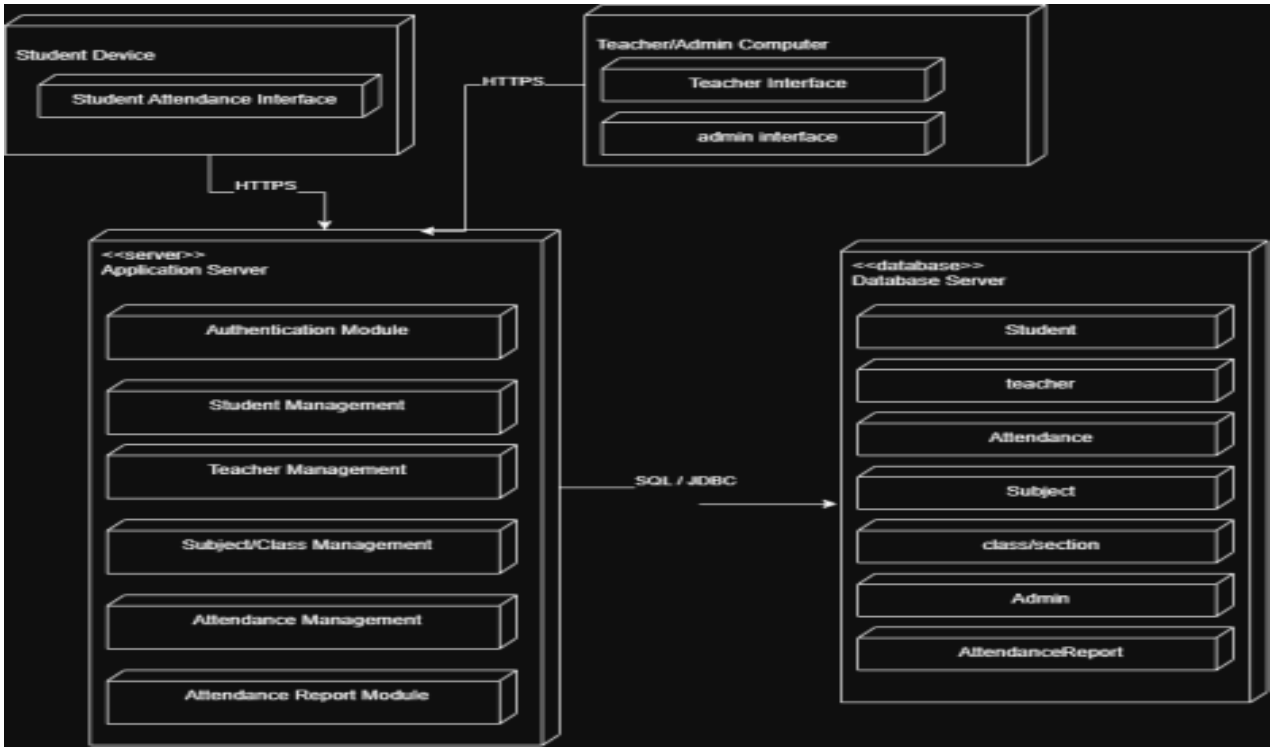
9.5 State Diagram

Shows the lifecycle of an attendance record from not recorded to finalized.



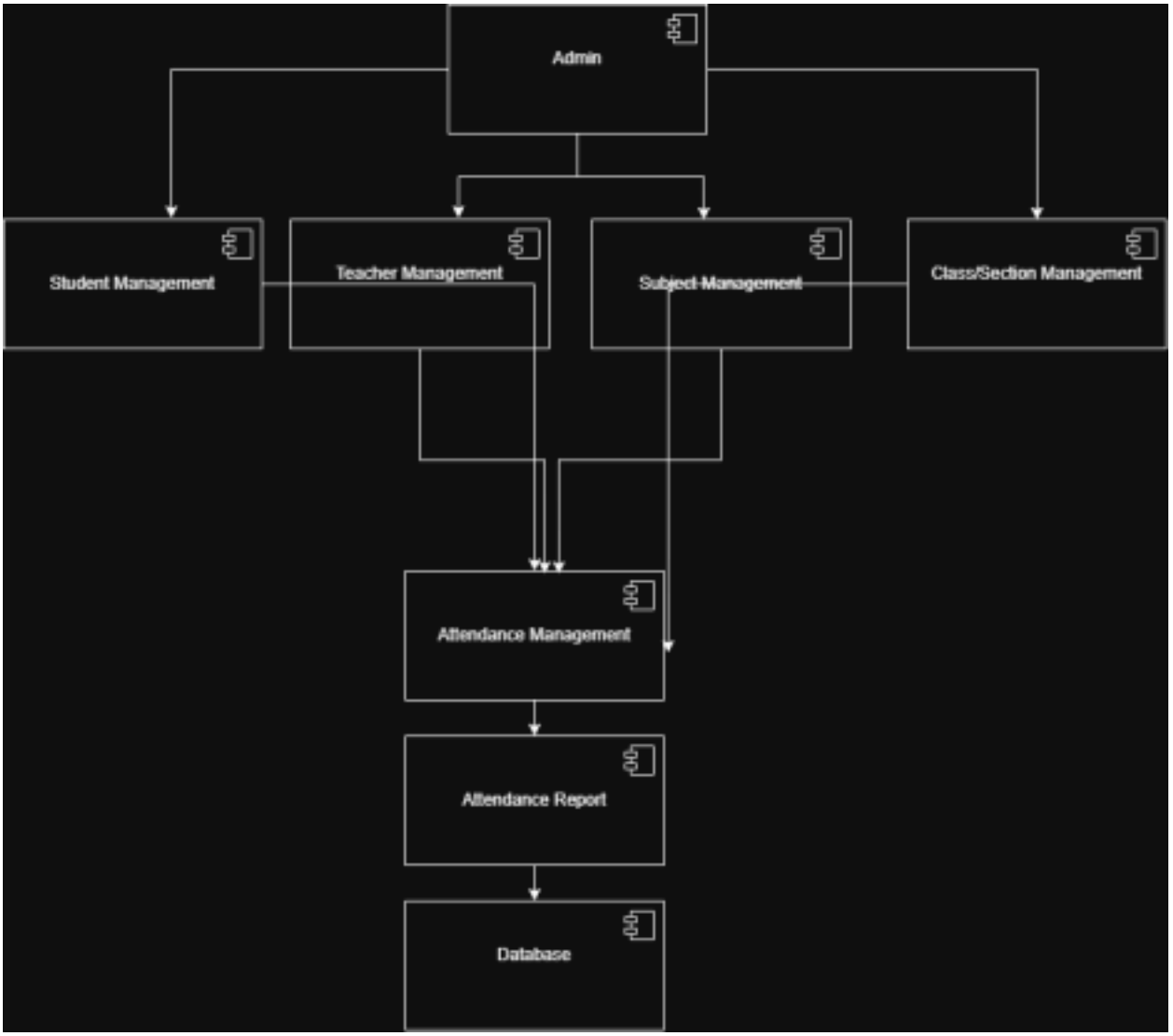
9.6 Deployment Diagram

Shows the physical deployment of client devices, application server, and database server.



9.7 Component Diagram

Shows the main software components and their relationships with the attendance database.



10. System Constraints

- Only authorized users can access protected features.
- Students cannot modify their own attendance.
- Duplicate attendance entries are not permitted for the same session.
- Attendance modification may be restricted to authorized users.
- The system depends on database and server availability.
- Network access is required for a web-based deployment.

11. Assumptions and Dependencies

Assumptions

- Each student has a unique student ID.
- Each teacher has a unique teacher ID.
- Students are assigned to classes.
- Teachers are assigned to subjects/classes.
- Users have valid login credentials.
- Attendance is recorded by authorized teachers or administrators.

Dependencies

- Database availability• Server availability
- Network connectivity
- Correct academic data
- Appropriate authentication mechanisms

12. Security Requirements

- **Authentication:** Users must authenticate before accessing protected functions.
- **Authorization:** Access shall depend on the user role.
- **Password Security:** Passwords shall not be stored as plain text.
- **Session Management:** Sessions should expire after an appropriate period of inactivity.
- **Input Validation:** User input shall be validated.
- **Database Security:** Database access shall be restricted to authorized application components.
- **Auditability:** Important attendance changes should record user identity and time.

13. Future Enhancements

- QR-code-based attendance
- RFID-based attendance
- Biometric attendance
- Face-recognition-based attendance
- Mobile application
- Parent/guardian portal
- Email, SMS, and push notifications
- Integration with college ERP systems
- Timetable integration
- Leave management
- Attendance analytics

14. Acceptance Criteria

- Users can log in according to their roles.
- Administrators can manage students, teachers, subjects, and classes.
- Teachers can record attendance.
- Duplicate attendance entries are prevented.
- Attendance percentages are calculated correctly.
- Students can view their own attendance.
- Authorized users can generate attendance reports.
- Unauthorized users cannot access restricted information.
- Attendance data is stored correctly.
- The UML diagrams represent the documented system requirements.

15. Conclusion

The Student Attendance Management System provides a centralized solution for recording, maintaining, calculating, and reporting student attendance. It replaces time-consuming manual processes with an organized digital workflow. The UML diagrams included in this document describe the system from functional, behavioral, structural, interaction, lifecycle, deployment, and component perspectives.

16. Reference

1.Software Requirements Specification (SRS) – Student Attendance Management System, Version 1.0, September 2026.

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3.diagrams.net (draw.io), *Diagramming and UML Modeling Tool*.

4.Ian Sommerville, *Software Engineering*, Pearson Education.

5.Roger S. Pressman and Bruce R. Maxim, *Software Engineering: A Practitioner's Approach*, McGraw-Hill Education.